ME597: Machine Learning (ML) Tutorial 2

Week 2

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Signal Processing – Time & Frequency Domain

PURDUE MECHANICAL ENGINEERING

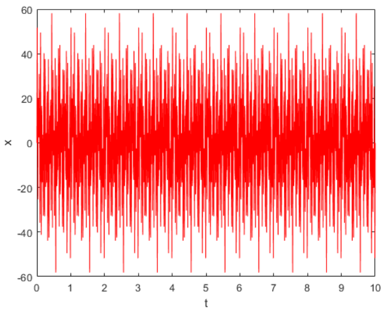
- Why we need to analyze the signal in frequency domain?

  - To observe characteristics of the signal that are not easy to see in the time domain.

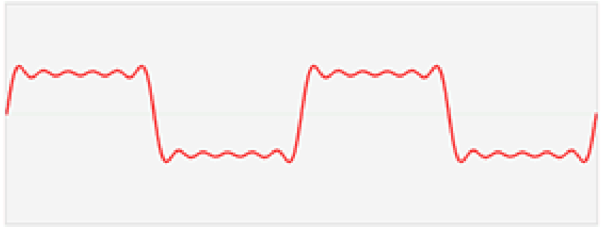
    (In general, motion of machines is implemented based on the rotation of motors)

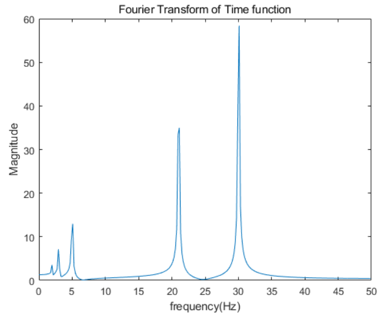
- Representative time-frequency transformation technique:

  - Fourier Transform (FT) – See Youtube "https://youtu.be/spUNpyF58BY"



Time Domain





Frequency Domain
(FT)

Fourier Transform – The Background

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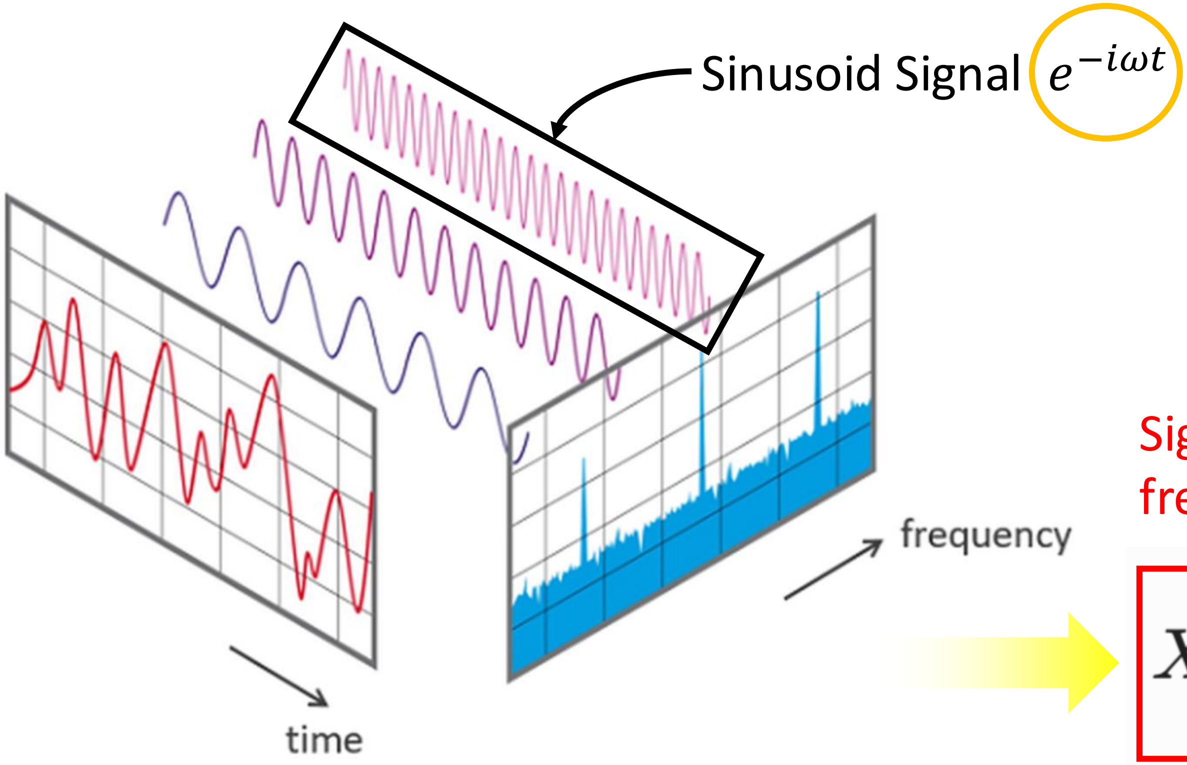
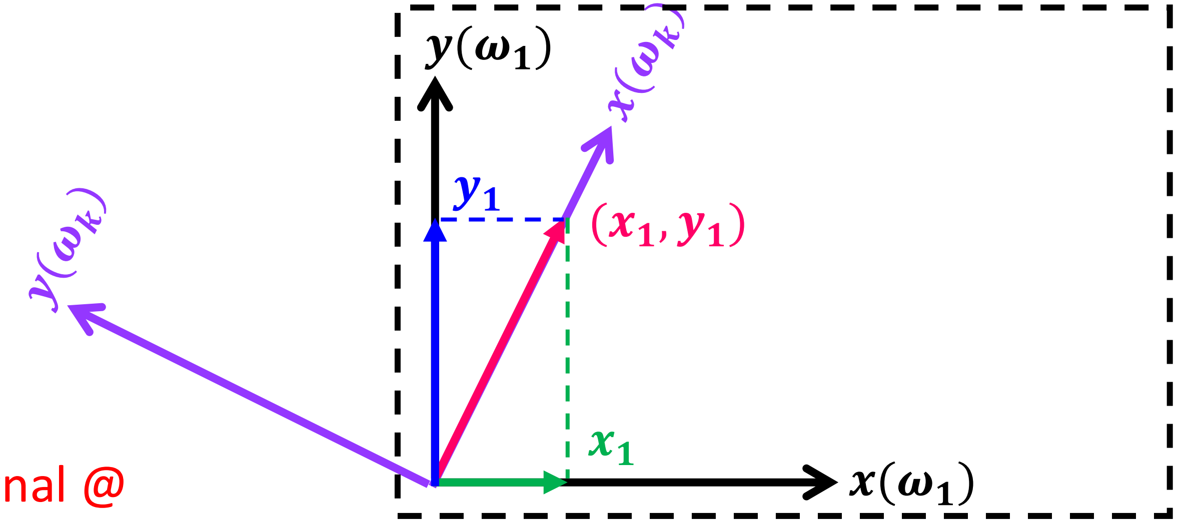
- Decomposition of Continuous Signal



Simplified Example



Signal @ frequency-domain

$$\boxed{X(\omega)} = \int_{-\infty}^{\infty} \boxed{x(t)}\, e^{-i\omega t}\, dt$$

Signal @ time-domain

$$e^{-i\omega t} = \cos \omega t + i \sin \omega t$$

Discrete Fourier Transform (DFT)

- Discrete Signal?

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt \quad \Rightarrow \quad X(k) = \sum_{n=0}^{N-1} x(\boxed{n}) e^{-j2\pi \boxed{kn}/N}$$

N sampled data points

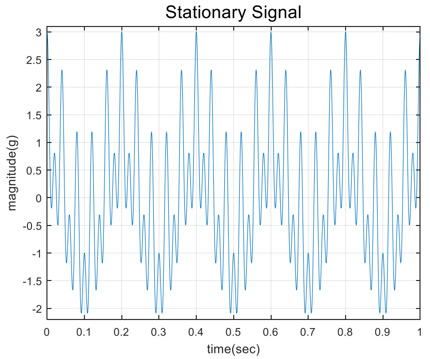
$$\boldsymbol{X(k)} = [\boldsymbol{x(0), x(1), x(2), \cdots x(k-1)}] \times \begin{array}{ccccc} \overbrace{\hphantom{xxxxxx}}^{k=0} & \overbrace{\hphantom{xxxxxx}}^{k=1} & \overbrace{\hphantom{xxxxxx}}^{k=2} & & \overbrace{\hphantom{xxxxxx}}^{k=N-1} \\ \end{array}$$

$$\begin{bmatrix} e^{-i2\pi 0 \frac{0}{N}} & e^{-i2\pi 0 \frac{1}{N}} & e^{-i2\pi 0 \frac{2}{N}} & \cdots & e^{-i2\pi 0 \frac{N-1}{N}} \\ e^{-i2\pi 1 \frac{0}{N}} & e^{-i2\pi 1 \frac{1}{N}} & e^{-i2\pi 1 \frac{2}{N}} & \cdots & e^{-i2\pi 1 \frac{N-1}{N}} \\ e^{-i2\pi 2 \frac{0}{N}} & e^{-i2\pi 2 \frac{1}{N}} & e^{-i2\pi 2 \frac{2}{N}} & \cdots & e^{-i2\pi 2 \frac{N-1}{N}} \\ \vdots & \vdots & \vdots & & \vdots \\ e^{-i2\pi (N-1) \frac{0}{N}} & e^{-i2\pi (N-1) \frac{1}{N}} & e^{-i2\pi (N-1) \frac{2}{N}} & \cdots & e^{-i2\pi (N-1) \frac{N-1}{N}} \end{bmatrix}$$

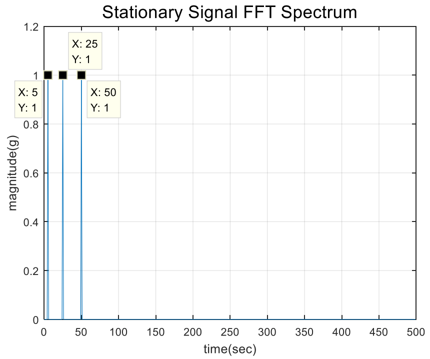
Signal Processing – Time & Frequency Domain

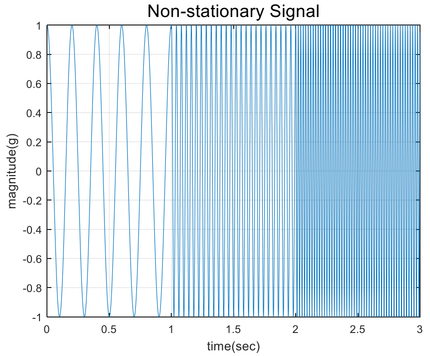
- Limitation of FFT

  - Stationary Signal : $\boldsymbol{cos}(\mathbf{5}\cdot\mathbf{2}\boldsymbol{\pi t}) + \boldsymbol{cos}(\mathbf{25}\cdot\mathbf{2}\boldsymbol{\pi t}) + \boldsymbol{cos}(\mathbf{50}\cdot\mathbf{2}\boldsymbol{\pi t})$
  - Non-stationary Signal : $\boldsymbol{vertex}\,[\boldsymbol{cos}(\mathbf{5}\cdot\mathbf{2}\boldsymbol{\pi t}),\ \boldsymbol{cos}(\mathbf{25}\cdot\mathbf{2}\boldsymbol{\pi t}),\ \boldsymbol{cos}(\mathbf{50}\cdot\mathbf{2}\boldsymbol{\pi t})\,]$

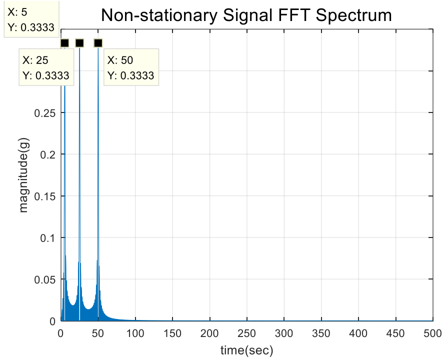


FFT





FFT



Not

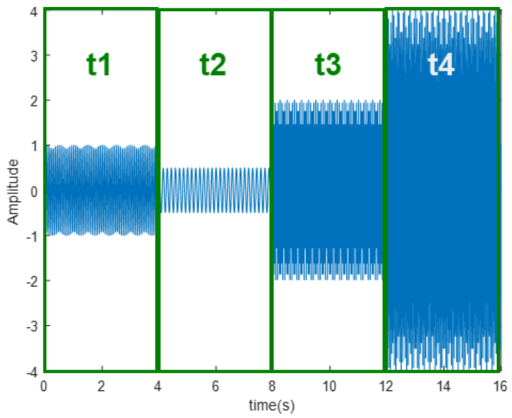
Distinguishable

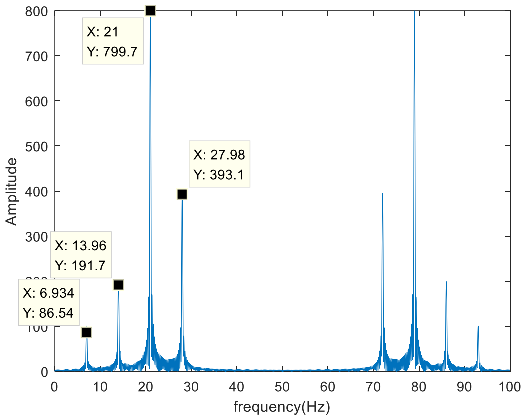
Signal Processing – Time & Frequency Domain

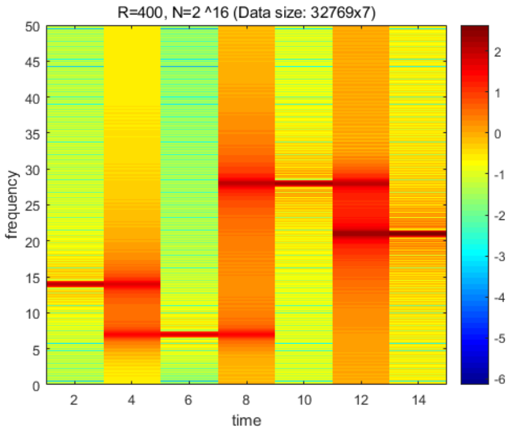
- FFT vs STFT (Short Time Fourier Transform)

  - STFT: Output a spectrogram that visualizes the frequency density (FFT) for each time segment

Non-stationary signal	FFT (Frequency Domain)	STFT (Spectrogram)
$t1 = 1 \times \cos(2\pi \times 14t)$ $t2 = 0.5 \times \cos(2\pi \times 7t)$ $t3 = 2 \times \cos(2\pi \times 28t)$ $t4 = 4 \times \cos(2\pi \times 21t)$	1-D (vector) data Inability to detect changes in frequency content over time	2-D (matrix) data Able to check frequency change over time







Signal Processing – Time & Frequency Domain

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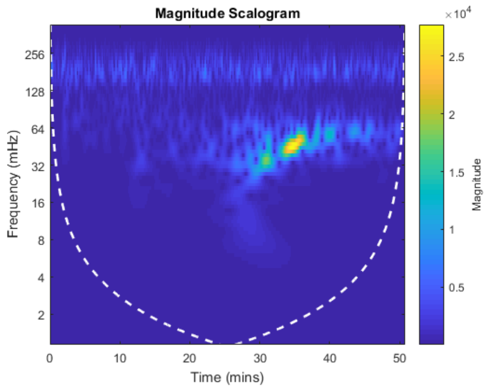
- WT (Wavelet Transform)

  - Able to adjust the frequency and time resolution by changing the Mother Wavelet Function ($\boldsymbol{\Psi}(\boldsymbol{X})$)

  (Basis function of FFT: Sine & Cosine / Basis function of WT: Mother wavelet)

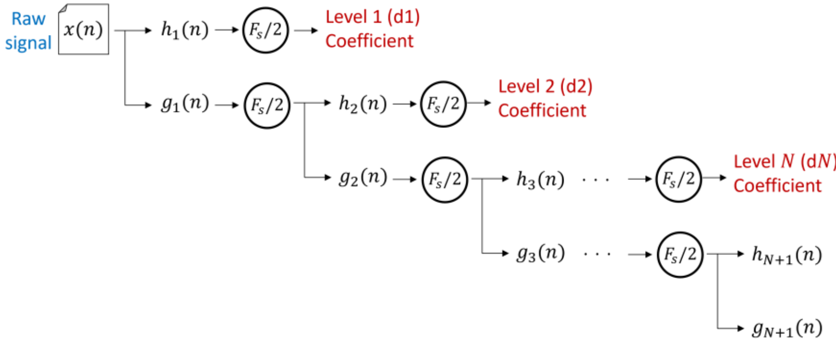


$$f(a,b) = \frac{1}{|a|^{1/2}} \int \Psi \frac{[x-b]}{a} f(x)dx$$

Mother Wavelet Function

a : Compression coefficient

b : Transfer coefficient



Representative Mother Wavelets

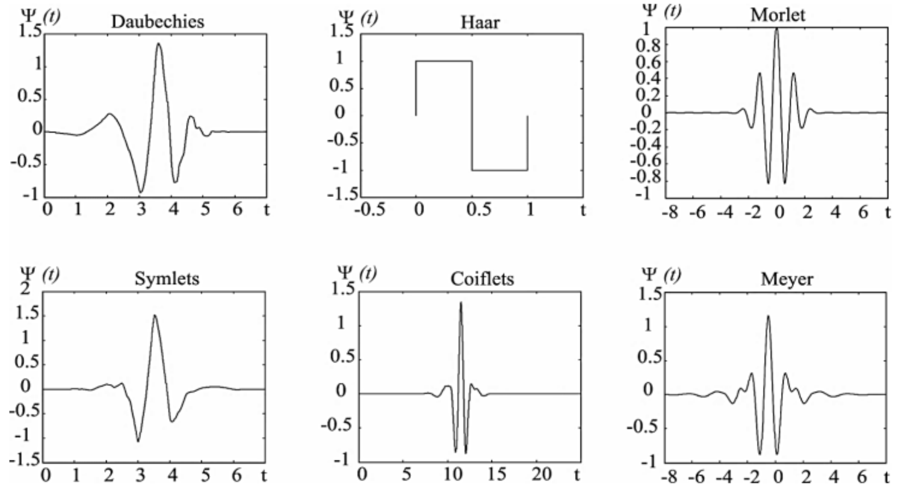


Fig. 9: Several families of wavelet transform.

[Ref.] Elaf Abed Saeed, et. al., "Series and Parallel Arc Fault Detection in Electrical Buildings Based on Discrete Wavelet Theory," IJEEE, Vol.17, 2(11) (2021)

Signal Processing – Time & Frequency Domain

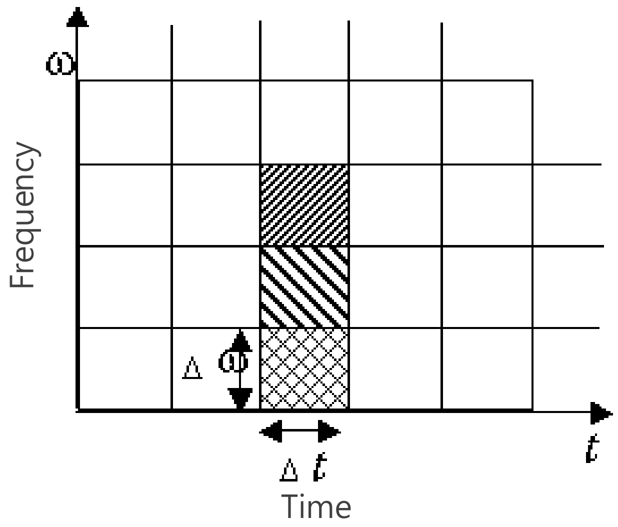
- STFT vs WT

  - STFT: Keep time resolution ($1/\Delta t$) and frequency resolution ($1/\Delta\omega$) constant (fixed by Window Function)

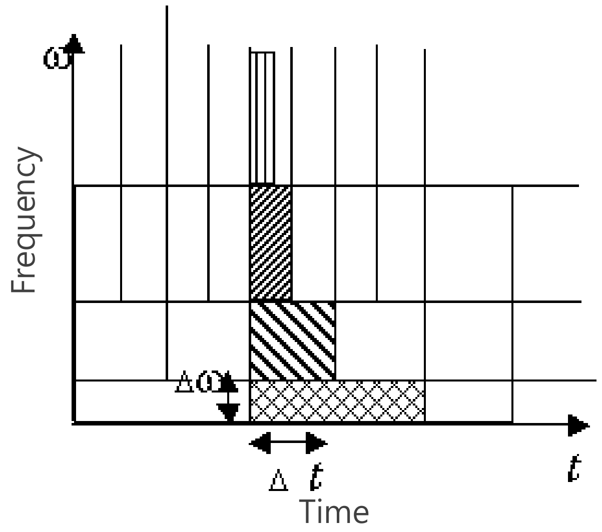
  - WT: Multi-resolution analysis of signals with complex frequency characteristics

    → In general cases, using STFT is enough to analyze the signal for machine monitoring

Short Time Fourier Transform



Wavelet Transform



Signal Processing – Time & Frequency Domain

- WT-based Signal Decomposition

  - Decompose the raw signal into multiple coefficient signals (time domain) by frequency range

    → Able to extract various time domain features from different frequency ranges

    → Key parameters: mother wavelet, number of coefficients